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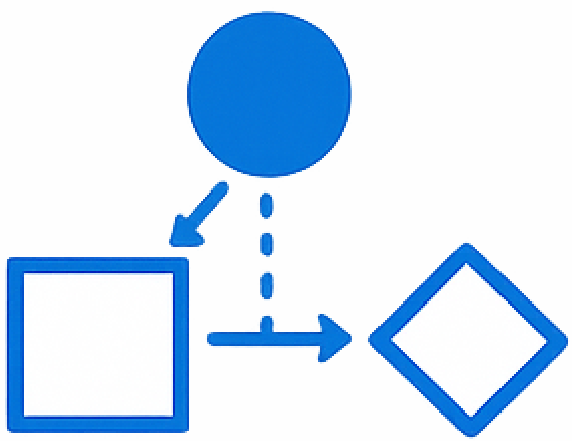

Assignment 2

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Qualitative Methods and Data Quality

Part I: Say-Do Gap Study and Part II: Radiology Data Quality Analysis

Date: March 2, 2026

Course: NDAB20010U Health Data and Interoperability

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AI Declaration

I declare that AI has only been used for grammatical correction.
AI has **not** been used to generate content, reasoning, or structure.

Contents

1 Part I: A Real-World Study of the Say-Do Gap in Modelling

This part analyses one weekday morning routine across *work-as-imagined*, *work-as-done*, and *work-as-recorded*, with explicit links between claims and evidence.

1.1 Deliverable 1: Description of the Routine Based on the Interview

Task Requirement (Deliverable 1) *Based on the interview, describe the routine and identify activities, their order, and participants involved.*

Scope and notation This Part I study uses a single-case self-study design with one participant (the author). The chosen routine is a weekday morning routine.

The interview was conducted on February 18, 2026 as a self-recorded, semi-structured interview with open questions. Following Guide to Health Informatics (2026a), this stage is treated as *manual knowledge acquisition*: it captures *knowledge-as-stated* and establishes a first *work-as-imagined* model of the routine (Guide to Health Informatics 2026; Zajac 2026). This placement is consistent with textbook Figure 27.1 from *Guide to Health Informatics*, where model acquisition precedes formal computer reasoning (Appendix B.4, Figure 6).

Identifier convention used in this report:

- **Axx** = *activity* IDs from the interview-based model (e.g., A01-A09).
- **Exx** = *event* IDs from the observed event log (e.g., E03, E10).
- **Dxx** = *delta* IDs in the comparison sheet (e.g., D01-D06).
- **Mx** = *method* IDs from the Part II DQ toolkit (e.g., M2, M10).

The model notation *DCR* denotes *Dynamic Condition Response* graph modelling.

The routine boundary is defined as start when the alarm sounds and end when the apartment door closes, based directly on transcript lines L21-L22 in Table 2. The baseline sequence (wake-up, hygiene, dressing/packing, breakfast, checks, departure) is grounded in lines L02-L12, and variation points (snooze, breakfast substitution, key search, occasional roommate coordination) are grounded in lines L14-L18. This line-level grounding makes the interview model auditable.

Routine boundary:

- Start: alarm sounds.
- End: participant leaves home (door closes).

Table 3 presents the interview-based activity model, and Figure 1 shows the corresponding *interview-based DCR* graph exported from DCR Graph. The graph was built from activities A01-A09 and used as the baseline representation for comparison against the observed event log in Deliverables 2 and 3.

Assumptions used for modelling:

- The model represents a weekday baseline (not weekend behavior).
- The participant is the only consistent human actor in the routine.
- Expected times are recall-based interview estimates and are treated as approximate.

These assumptions delimit model scope and support transparent comparison between *knowledge-as-stated* and observed evidence in Deliverables 2 and 3.

Interview transcript evidence The full line-numbered transcript excerpt used as evidence for Deliverables 1-4 is provided in Appendix A, Table 2.

Interview-based activity model The full interview-based activity table and DCR figure are provided in Appendix A, Table 3 and Figure 1. These appendixed artefacts are the baseline *work-as-imagined* model used for Deliverables 2 and 3.

1.2 Deliverable 2: Event Log Representing the Observed Routine

Task Requirement (Deliverable 2) *Create an event log representing the observed routine as a .csv file with timestamps, event labels, and relevant context/notes.*

The observation was conducted on February 20, 2026 (06:45-07:30), two days after the interview on February 18, 2026. This separation supports comparison between interview elicitation and situated execution.

As described in Chapter 27 and the course methods slides, *observation* complements *interviews* by capturing situated judgement and practical adaptations that are often missing from recollection-based accounts (Guide to Health Informatics 2026; Zajac 2026). To make data collection explicit and reproducible, a short observation guide was used (Appendix A, Table 1). Notes were split into *descriptive notes* and *reflexive notes*, and only descriptive notes were transcribed into the event log (Zajac 2026).

The full machine-readable event log is submitted as a separate CSV artifact with timestamps, event labels, and context notes. This file is the *work-as-recorded* layer of the study and the empirical basis for the *observed DCR* comparison in Deliverables 3-4.

The submitted CSV uses the following fields: `case_id`, `event_id`, `timestamp` (ISO datetime), `event_label`, `actor`, `tool`, `context_note`, and `related_activity_id` (A01-A09 or NA).

Table 4 in Appendix A provides the observed event-log excerpt with activity references. Relative to the transcript-based baseline (Table 2, lines L02-L12 and L14-L18), the key comparison anchors are E03 (extra coordination), E09 (order shift), and E10 (workaround).

The observation confirms the same overall routine boundary as Deliverable 1, but adds practical variation and coordination not present in the interview baseline. In particular, one extra coordination event (E03), one order change (E09 relative to A04/A05), and one explicit workaround (E10) are central for the delta sheet in Deliverable 3. These events represent *work-as-done* under actual constraints.

1.3 Deliverable 3: Delta Sheet Summarising Differences Between Stated and Observed Practice

Task Requirement (Deliverable 3) *Compare Deliverable 1 and Deliverable 2 and produce a delta sheet categorised into missing events, extra events, different order, timing distortions, workarounds/coordination steps, and relabeling/ambiguity.*

The comparison maps interview activities (A01-A09) to observed events in Deliverable 2. A delta was registered when at least one of the following differed: event presence, order, timing, workaround behavior, or interpretation of activity labels. Methodologically, this follows Chapter 27's logic of comparing observed process traces with a pre-existing process description to identify frequent exceptions and model limits (Guide to Health Informatics 2026).

Table 5 in Appendix A summarises the full delta comparison between stated and observed routine. Key deltas are:

- *Missing event*: no distinct seated breakfast-and-schedule-review step in observation.
- *Extra event*: additional roommate coordination event appears only in observed data.
- *Different order*: bag packing occurs after breakfast preparation begins.
- *Timing distortion*: hygiene block starts later than stated baseline.
- *Workaround/coordination step*: missing lock triggers ad hoc borrowing via messaging.
- *Relabeling/ambiguity*: "breakfast" shifts from sit-down meal to banana plus coffee-to-go.

All six required categories are documented in Table 5: missing events, extra events, different order, timing distortions, workarounds/coordination steps, and relabeling/ambiguity.

Overall, the *interview-based DCR* (Deliverable 1) captures *knowledge-as-stated* and *work-as-imagined*, while the observed event sequence (*observed DCR* evidence) captures *knowledge-as-discovered*, *work-as-done*, and *work-as-recorded*. The delta sheet therefore provides traceable evidence of the *say-do gap*.

1.4 Deliverable 4: Reflections on the Modelling Process

Task Requirement (Deliverable 4) *Reflect on six modelling questions: data missed/misrepresented, activity definition differences, event-log constraints, automation potential, variations revealed by observation, and why qualitative discovery matters for process modelling.*

1. **Question 1:** *What kinds of data were missed or misrepresented in the interview-based model and why?* **Answer:** The interview model captured routine intent, but missed situated details that appeared in observation: micro-coordination (E03), recovery behavior (E10), and omission of the planned seated breakfast-and-schedule step. This pattern is expected in *manual knowledge acquisition*: *interviews* primarily capture recalled intent, while *observation* captures situated adaptation (Guide to Health Informatics 2026; Zajac 2026). In this case, the interview model therefore overestimated routine linearity.

2. **Question 2:** *What counts as an activity in each model, and how did this differ?* **Answer:** In Deliverable 1, the *interview-based DCR* uses intention-level activities (A01-A09). In Deliverable 2, the event log (the empirical basis for the *observed DCR*) uses timestamped observable actions. This granularity mismatch means one DCR activity can map to multiple logged events. The same routine can therefore appear stable at intention level but variable at event level.

3. **Question 3:** *How did the event log constrain what could be modelled?* **Answer:** The event log supports strong analysis of order, timing, and deviations (*work-as-recorded*), but it captures motives and judgement only indirectly. Without contextual notes, causal interpretation of why a deviation occurred remains weak. This is why descriptive/reflexive note separation was retained in the method and documented in Appendix A, Table 1.

4. **Question 4:** *Do you think that the event log could be captured automatically by a system? Why or why not?* **Answer:** Automatic capture is strong for digital traces (alarm, app checks, timestamps), but weak for tacit coordination, local workarounds, and searches for missing items. In Chapter 27 terms, automated traces should be combined with *qualitative discovery* methods when the task includes social and contextual adaptation (Guide to Health Informatics 2026; Zajac 2026).

5. **Question 5:** *How did the observation help uncover variations or exceptions that were not mentioned in the interview?* **Answer:** Observation revealed concrete *exceptions*: changed order (E07 before E09), omitted planned step (A06 not distinct), and workaround behavior (E10). Applied to this case, Chapter 27 implies that process evidence should not only test conformance but also explain recurring exceptions that may be locally rational (Guide to Health Informatics 2026).

6. **Question 6:** *Why is it relevant to consider qualitative methods of discovery (e.g., observations) when building process models?* **Answer:** For this routine, model quality improved by triangulating *knowledge-as-stated* (interview + transcript), *knowledge-as-discovered* (observation), and *work-as-recorded* (event log). The resulting model is more realistic than any single method alone, consistent with Chapter 27's iterative *knowledge acquisition* approach (Guide to Health Informatics 2026).

2 Part II: Radiology Case Study — Data Quality, Conformance, and the Limits of Logs

This part synthesises notebook evidence (Deliverable 5) and the accompanying Part II reflection section (Deliverable 6).

2.1 Deliverable 5: Notebook Submission

Task Requirement (Deliverable 5) *Submit the completed notebook `assignment.ipynb` with the Part II workflow and method execution.*

The completed notebook is submitted as a separate artifact (`assignment.ipynb`), as required by the assignment brief.

The notebook contains the full Part II workflow (Parts 0–7): initial exploration, method selection and execution, custom *DQ dimension* design, *five-facet* synthesis (*Data, Source, System, Task, Human*), and *invisible-work* analysis.

Deliverable 6 and Deliverable 7 both interpret these outputs as *work-as-recorded* evidence, with different wording alignment (assignment brief vs notebook template).

2.2 Deliverable 6: Part II Reflection Questions

A Supporting Tables for Part I

This appendix contains supplementary Part I material referenced from Deliverables 1–4.

Table 1: Observation guide used during Deliverable 2

Focus	Observation prompt	Evidence target
Sequence and boundary	Which event marks start/end, and what is the observed order of actions?	Event IDs, timestamps, sequence notes
Tools and artefacts	Which tools/resources are used at each step?	Tool field, context notes
Interactions and coordination	Does any interaction with other people occur?	Actor field, coordination notes
Deviations from interview baseline	Which actions are skipped, added, reordered, or relabeled relative to A01-A09?	Related activity ID + delta notes
Adaptation and recovery	Which workaround or exception handling occurs under time/resource pressure?	Context notes for exception events

Source: Zajac (2026) and Guide to Health Informatics (2026a): interview + observation + process-tracing principles adapted to this routine study.
Note: Field notes were split into descriptive and reflexive notes; only descriptive notes were transcribed into the submitted CSV event log.

A.1 Core Deliverable Artefacts (Part I)

Table 2: Line-numbered self-interview transcript (conducted February 18, 2026)

Line	Transcript
L01	Interview question: Please describe your normal weekday morning routine from when you wake up until you leave home.
L02	Participant: My alarm goes off at 06:45, and I usually get up right away after stopping it.
L03	Participant: I first drink a glass of water, then I go to the bathroom.
L04	Participant: I normally take a quick shower, around ten to twelve minutes.
L05	Participant: After that I get dressed and prepare my bag for the day.
L06	Participant: I usually pack my laptop, charger, student ID card, wallet, and keys.
L07	Participant: Then I make coffee and toast.
L08	Participant: I normally sit down, eat breakfast, and check my calendar and transport timing on my phone.
L09	Participant: After breakfast I brush my teeth and do a final mirror check.
L10	Participant: I check the weather so I can choose the right jacket.
L11	Participant: Before leaving, I do a final essentials check: phone, wallet, keys, and card.
L12	Participant: I aim to leave at around 07:30 to catch transport to campus.
L13	Interview question: Which steps vary the most from day to day?
L14	Participant: Sometimes I snooze once if I am tired.
L15	Participant: Breakfast can change; if I am in a rush, I might just eat fruit instead of toast.
L16	Participant: Occasionally I spend a minute looking for keys if I did not put them in the same place.
L17	Interview question: Are other people typically involved in this routine?
L18	Participant: Mostly no, I do it alone, but once in a while I text my roommate if I need a shared item.
L19	Interview question: Which tools or systems are central in the routine?
L20	Participant: My phone alarm, calendar, weather app, the coffee machine, toaster, and my backpack.
L21	Interview question: How do you define start and end of this routine?
L22	Participant: It starts when the alarm sounds and ends when I close the apartment door behind me.

Table 3: Interview-based activity model (work-as-imagined / knowledge-as-stated)

ID	Activity label	Actor	Order	Expected time	Main tools/resources
A01	Stop alarm and orient to time	Participant	1	06:45-06:47	Smartphone alarm
A02	Drink water	Participant	2	06:47-06:50	Glass, tap
A03	Toilet and shower	Participant	3	06:50-07:02	Bathroom, shower
A04	Dress and pack bag	Participant	4	07:02-07:12	Wardrobe, backpack
A05	Prepare breakfast and coffee	Participant	5	07:12-07:20	Coffee machine, toaster
A06	Eat breakfast and review schedule	Participant	6	07:20-07:25	Phone calendar/transport apps
A07	Brush teeth and final grooming	Participant	7	07:25-07:28	Toothbrush, mirror
A08	Check weather and essentials	Participant	8	07:28-07:30	Weather app, keys, wallet, ID
A09	Leave apartment for commute	Participant	9	07:30	Front door, bag

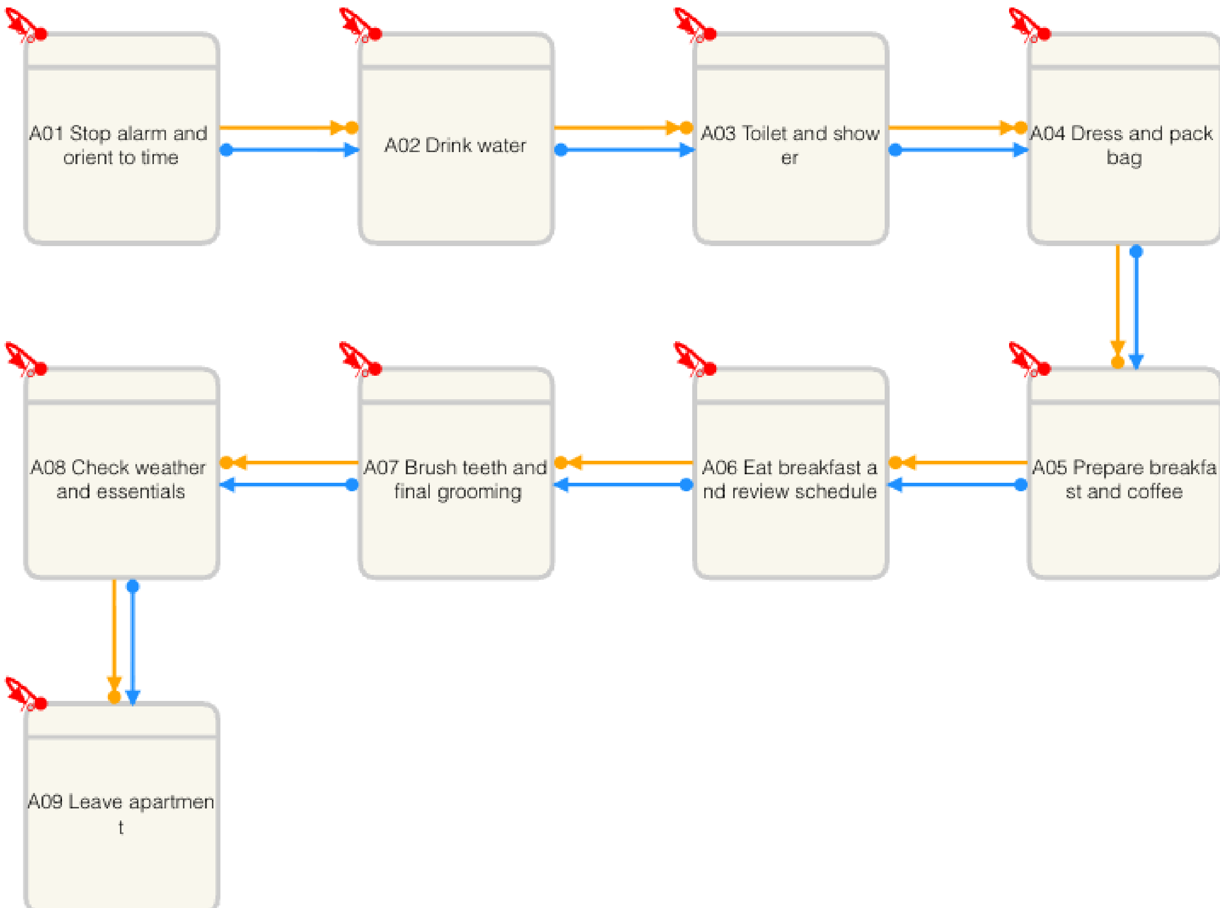

Figure 1: Interview-based DCR graph for the morning routine (A01-A09)

Source: Own DCR Graph export (February 2026), built from transcript evidence in Table 2 (lines L02-L12, L14-L18, L21-L22).

Note: Used as baseline work-as-imagined model for comparison with observed events in Deliverables 2 and 3.

Table 4: Observed event-log excerpt (February 20, 2026)

Event	Timestamp	Event label	Tool	Context note	Ref.
E01	2026-02-20 06:45	Alarm rings and snooze	Smartphone alarm	One snooze before wake-up.	A01
E03	2026-02-20 06:52	Reply to roommate message	Messaging app	Extra micro-coordination event.	NA
E05	2026-02-20 06:57	Toilet and shower	Bathroom	Hygiene block started later than interview expectation.	A03
E07	2026-02-20 07:12	Start coffee and toast	Coffee machine, toaster	Breakfast preparation begins.	A05
E08	2026-02-20 07:16	Switch breakfast to banana	Kitchen	Breakfast adapted due to no bread.	A05
E09	2026-02-20 07:18	Pack bag	Backpack	Packing occurs after breakfast prep starts.	A04
E10	2026-02-20 07:21	Find keys and borrow lock	Keys, messaging app	Workaround to solve missing item.	A08
E14	2026-02-20 07:30	Leave apartment	Front door	End of observed routine boundary.	A09

Table 5: Delta sheet between interview model and observed routine

Delta ID	Category	Interview ref.	Observed ref.	Difference description	Implication
D01	Missing events	A06	No direct event	No distinct seated breakfast-and-schedule-review event occurred.	Interview model overstates routine regularity.
D02	Extra events	None in baseline	E03 (06:52)	Short roommate-message coordination appears in observed routine only.	Reveals practical coordination load.
D03	Different order	A04 before A05	E07 then E09	Bag packing happened after breakfast preparation started.	Ordering constraints are weaker than stated.
D04	Timing distortions	A03 expected 06:50-07:02	E05 start 06:57	Hygiene block shifted later due to wake-up delays.	Time windows in interview are optimistic.
D05	Workarounds/coordination steps	A08 final check	E10 (07:21)	Missing lock triggered ad hoc borrowing via message.	Recovery actions are underrepresented in interview model.
D06	Relabeling/ambiguity	A05/A06	E08, E13	“Breakfast” changed from sit-down meal to banana plus coffee-to-go.	Label definitions need operational clarity.

B Supporting Tables and Code for Part II

This section includes only the relevant outputs from the completed Deliverable 5 notebook (`assignment.ipynb`): selected method evidence, key visualisations, custom DQ-dimension results, and invisible-work comparison material.

Table 6: Part II log profile from notebook Part 1 exploration

Metric	Result
Total events	150
Unique cases	17
Time range covered	2026-01-12 07:45 to 2026-01-13 08:46
Unique logged activities	12
Columns with strongest missingness	`study_uid` (134), `series_count` (134), `instance_count` (134), `icd11_code` (131), `report_id` (113), `modality` (101), `notes` (73)

Table 7: Selected methods run in notebook Part 4

Method	DQ dimension	Primary facets	RQ link	Selection rationale
M2	Completeness	Data, Source, System	RQ1–RQ3	Baseline quality gate before downstream interpretation.
M6	Representativity	Data, Task, Human	RQ1, RQ3	Checks urgency and temporal balance of case coverage.
M8	Timeliness conformance	Task, System, Human	RQ2	Tests safety-critical communication behavior directly.
M10	Timeliness + credibility	Data, System, Task	RQ1	Quantifies TAT and sensitivity to batch logging.
M11	Workload representativity	Human, Task, System	RQ1, RQ3	Reveals concentration across actors/roles.

Table 8: Key quantitative outputs from selected methods

Method	Key output	Relevance for Deliverable 6 reflection
M2	Context-field missingness is high (e.g., `study_uid` 134/150; `modality` 101/150).	Shows what can and cannot be trusted for workflow and AI-oriented interpretations.
M6	Urgency skew (15 stat/urgent, 2 routine) and shift imbalance (Day 11, Night 6, Evening 0).	Demonstrates representativity limits for RQ1 and especially RQ3 generalisation.
M8	Critical communication: 16 pass, 1 fail (`CXR-004` missing communication event).	Direct patient-safety signal tied to RQ2 and conformance assumptions.
M10	Mean total TAT = 147.93 min raw vs 198.50 min clean; 6/17 cases flagged as batch-logged (35.29%).	Shows strong sensitivity of performance conclusions (RQ1) to timestamp credibility.
M11	Events per role: radiographer 64, system 31, senior 31, clinician 17, resident 7.	Quantifies workload concentration and role imbalance relevant to RQ1 and RQ3.

B.1 Notebook Visualisations (Relevant Outputs)

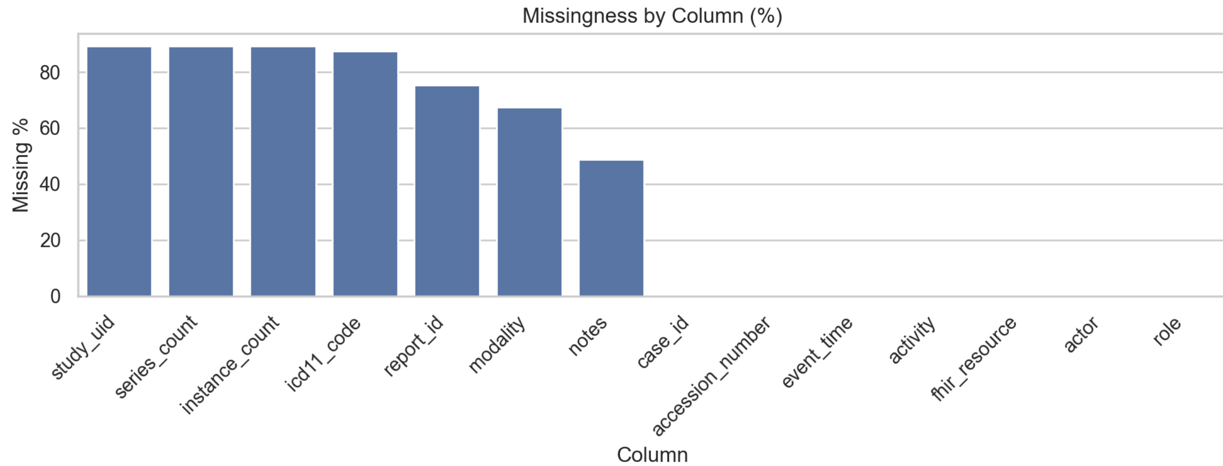

Figure 2: Method 2 output: missingness profile by column

Source: Generated from Deliverable 5 notebook output using `radiology_event_log.csv`.

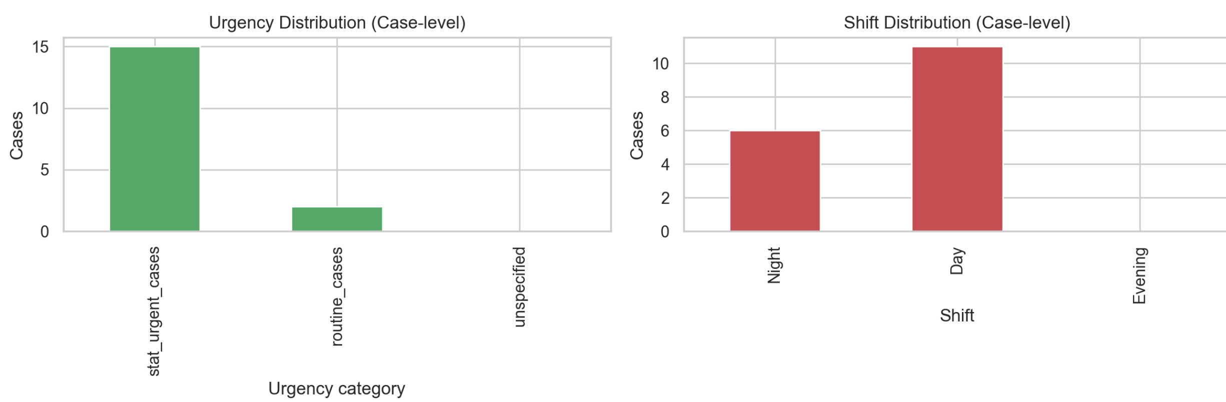

Figure 3: Method 6 output: urgency and shift representativity

Source: Generated from Deliverable 5 notebook output using `radiology_event_log.csv`.

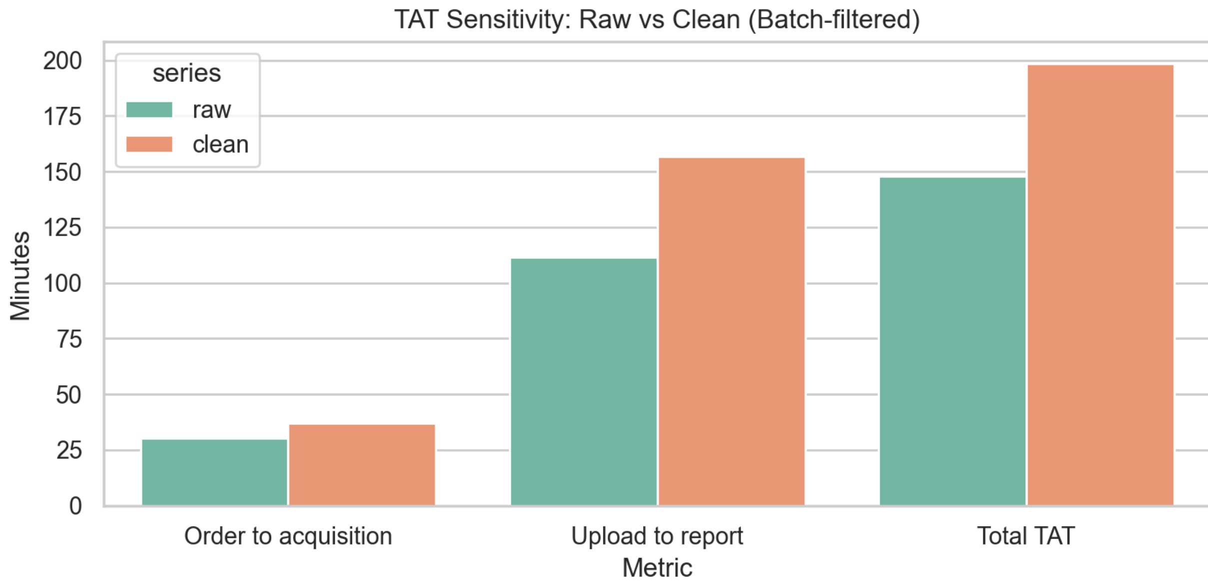

Figure 4: Method 10 output: turnaround-time sensitivity (raw vs batch-filtered)

Source: Generated from Deliverable 5 notebook output using radiology_event_log.csv.

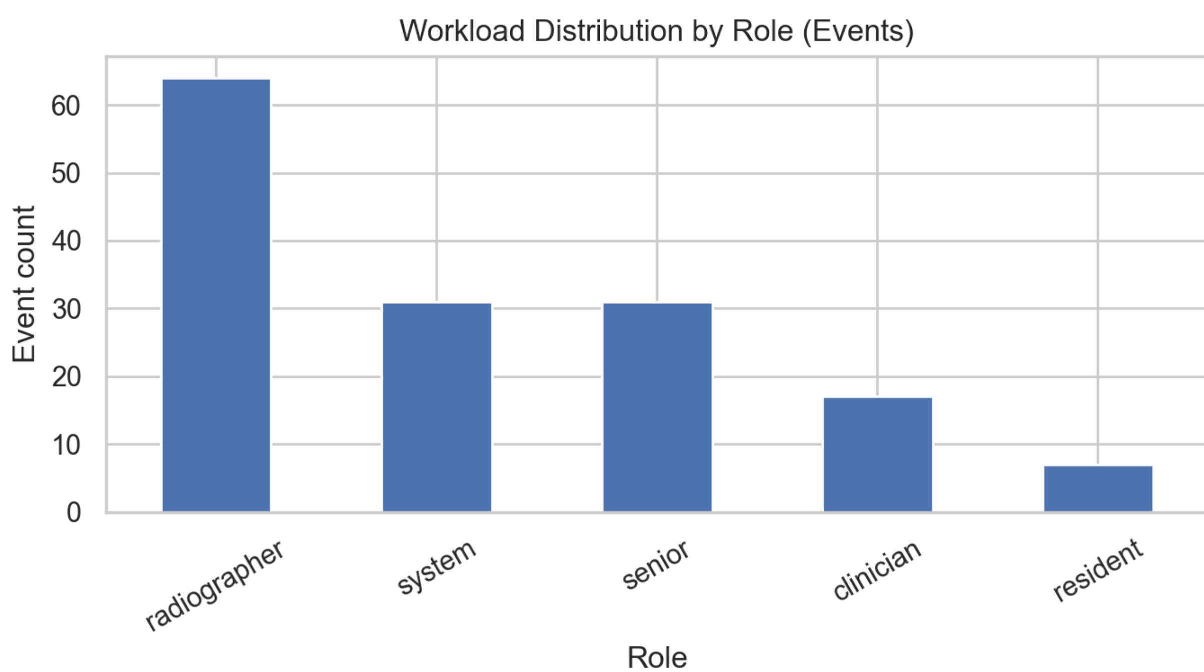

Figure 5: Method 11 output: workload distribution by role

Source: Generated from Deliverable 5 notebook output using radiology_event_log.csv.

B.2 Custom DQ Dimension (Notebook Part 5)

Table 9: Custom DQ dimension result summary: operational authenticity

Metric	Result
Total evaluated cases	17
Flagged cases (medium/high risk)	1
Flagged case rate	5.88%
High-risk cases	1
Flagged case details	`TEST-001`: unexpected case ID format; training/test marker in notes; administrative actor involvement; extremely short trace (1 event)

Table 10: Facet mapping for custom dimension: operational authenticity

Facet	Involvement	Justification
Data	++	Case identifiers and note text contain direct contamination signals (test/training traces).
Source	++	Source governance determines whether non-production traces enter the analysis dataset.
System	+	Logging architecture does not fully separate test and production provenance.
Task	+	KPI and triage conclusions become biased when non-clinical traces are included.
Human	+	Human setup/maintenance actions can introduce non-operational events.

B.3 Invisible Work Material (Notebook Part 7)

Table 11: Activities logged in the event log (12 unique activities)

Logged activity set A	Logged activity set B
Case selected	Order placed
Critical communication	Patient selected from worklist
Image acquisition start	Preliminary report created
Image acquisition end	Report distributed
Images uploaded to PACS	Report finalised
Order accepted	Report validated and approved

Table 12: Invisible-work activities from workflow diagram not represented explicitly in log

Activity from diagram	Why not logged?	Facet	Consequence	Capture option
Radiologist-to-radiologist consultation	Mostly verbal/ad hoc interaction outside structured capture	Human, Task	Collaborative interpretation work is under-represented	Optional lightweight consult event
Clarification call to ordering clinician	Usually done by phone outside RIS/PACS event schema	System, Source	Communication-delay attribution becomes incomplete	Linked communication event type
Informal reprioritisation and handoff	Queue coordination is locally negotiated rather than formalised	Task, Human	Bottleneck and fairness analyses miss practical prioritisation logic	Priority-change event with rationale
Cross-team coordination on patient readiness	Interdepartmental coordination spans systems and roles	Source, System, Task	Waiting-time drivers become invisible in case timeline analysis	Interdepartment status event

B.4 Textbook Figures Referenced in the Report

The figures below are included to support direct references in Part I and Part II. To separate them from assignment-produced figures, each caption explicitly preserves the textbook figure number.

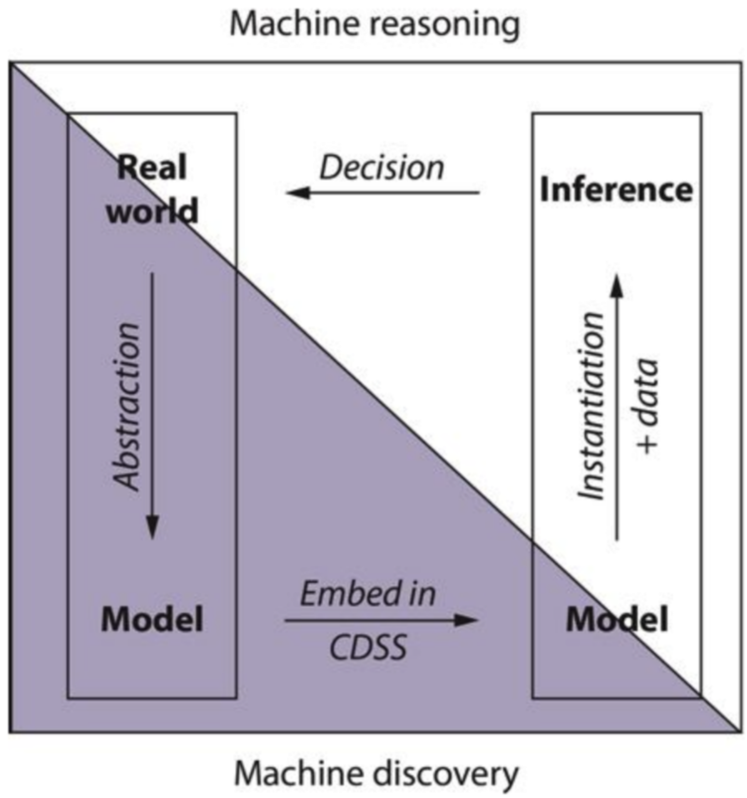

Figure 27.1 Computer reasoning depends first on acquiring models that record our understanding of how the world works and then applying those models in a decision support system to arrive at conclusions shaped by the data specific to a given decision task.

Figure 6: Textbook Figure 27.1 from *Guide to Health Informatics*: computer reasoning cycle from model construction through use

Source: Guide to Health Informatics (2026a). Reproduced from course compendium reading.

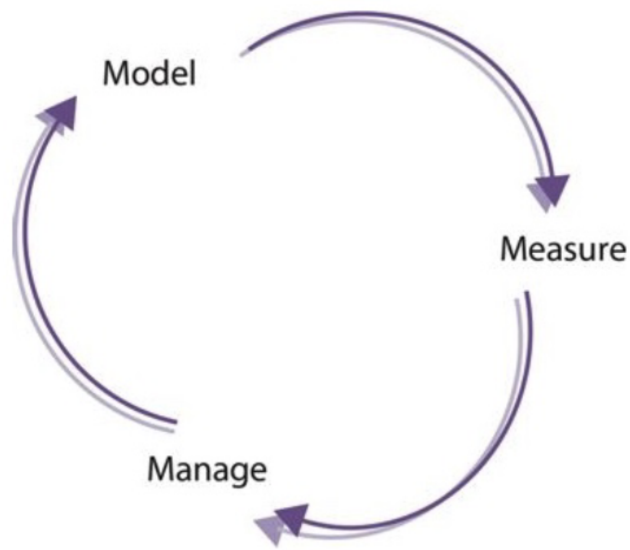

Figure 9.1 The model-measure-manage cycle. When the outcome of a management decision is fed back into another cycle, a feedback control loop is formed.

Figure 7: Textbook Figure 9.1 from *Guide to Health Informatics*: model-measure-manage cycle

Source: Guide to Health Informatics (2026b). Reproduced from course compendium reading.

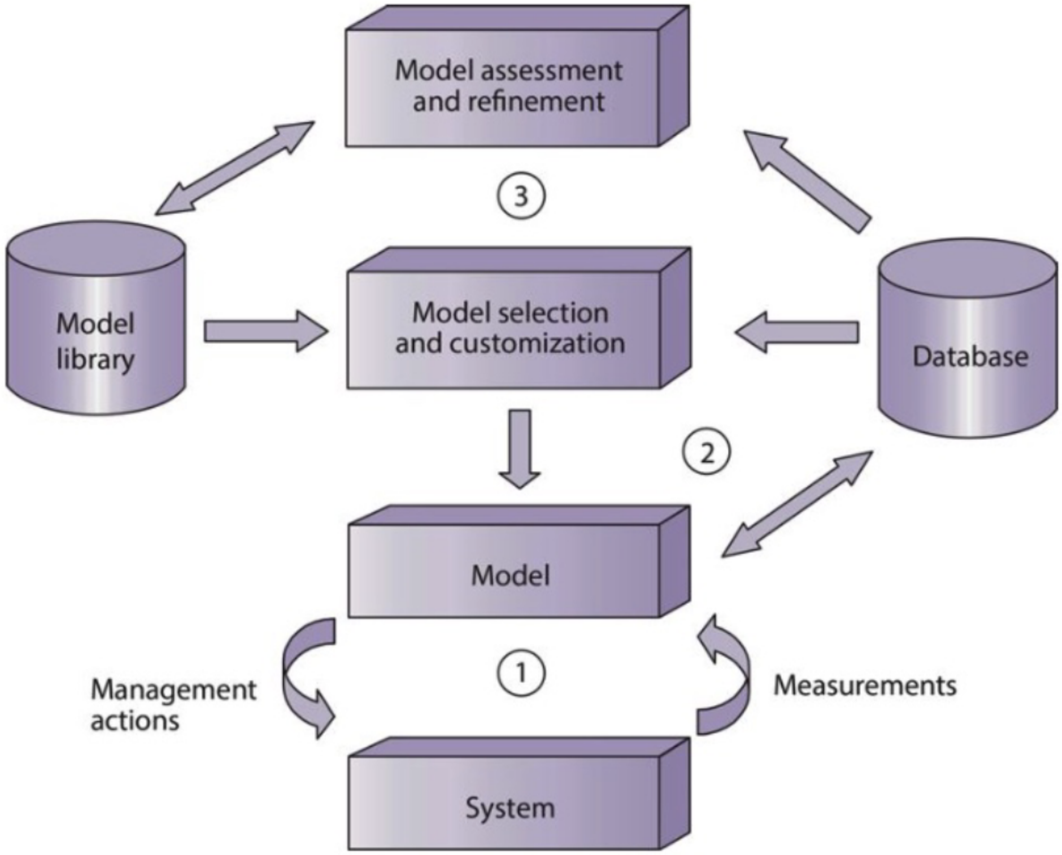

Figure 9.2 The three-loop model. Three separate information cycles interact within the health system. The first and second cycles select and apply models to the management of specific systems. The third information cycle tries to assess the effectiveness of these decisions and improve the models accordingly.

Figure 8: Textbook Figure 9.2 from *Guide to Health Informatics*: three-loop model

Source: Guide to Health Informatics (2026b). Reproduced from course compendium reading.

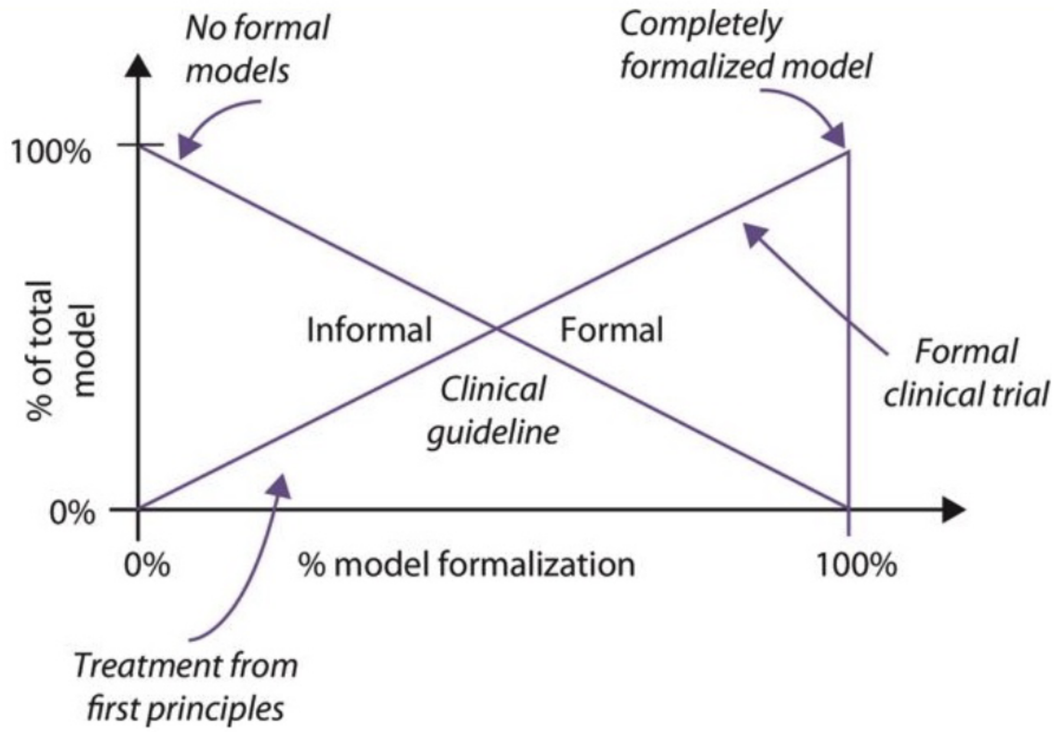

Figure 9.3 There is a continuum of possible formalization of information systems. Formal and informal models combine together to cover the total modelling needs of a given problem. The degree of formalization depends on the type of problem being addressed.

Figure 9: Textbook Figure 9.3 from *Guide to Health Informatics*: formalization continuum

Source: Guide to Health Informatics (2026b). Reproduced from course compendium reading.

References

Guide to Health Informatics (Course Compendium) (2026a). *Guide to Health Informatics: Selected Chapter 27*. PDF. Course reading PDF provided in NDAB20010U.

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